

Editorial

Partial Least Squares Structural Equation Modeling: Rigorous Applications, Better Results and Higher Acceptance

Wold's (1982, 1974) and Lohmöller's (1989) partial least squares structural equation modeling (PLS-SEM) approach has enjoyed increasing popularity as a key multivariate analysis method in various research disciplines such as accounting (Lee et al., 2011), international marketing (Henseler et al., 2009), management information systems (Ringle et al., 2012), marketing (Hair et al., 2012c) and operations management (Peng and Lai, 2012). Similarly, strategic management research has recognized PLS-SEM's flexibility regarding the handling of various modeling problems (e.g., Hulland, 1999). However, according to Hair et al.'s (2012b) recent review of PLS-SEM use in the strategic management field, the method's dissemination is not as widespread as in other fields.

This lower dissemination suggests that PLS-SEM use in strategic management presents numerous application opportunities. More often than not, researchers do not pay attention to data quality requirements (e.g., with regard to sample size or treatment of missing values) and disregard important evaluation criteria (especially when assessing formative measurement models), sometimes even misapplying them (e.g., by using reflective measurement model assessment procedures on formative measurement models). Most studies do not provide sufficient information to enable readers to replicate the results and fully assess a study's quality. Consequently, Hair et al. (2012c; p. 430) conclusion in the field of marketing also seems to hold for the strategic management discipline: "While offering many beneficial properties, PLS-SEM's 'soft assumptions' should not be taken as *carte blanche* to disregard standard psychometric assessment techniques." Furthermore, researchers do not make use of the entire toolbox that PLS-SEM offers. Methodological advances (Esposito Vinzi et al., 2010; Hair et al., 2013) provide researchers much more flexibility in modeling relationships and thus allow for a more nuanced testing of theoretical concepts. Amongst others, these advances include confirmatory tetrad analysis (CTA-PLS) to empirically assess the measurement model type (i.e., formative or reflective; Gudergan et al., 2008), importance-performance matrix analysis (IPMA) of PLS-SEM results (e.g., Höck et al., 2010; Rigdon et al., 2011; Völckner et al., 2010), approaches to assess hierarchical component models (e.g., Becker et al., 2012; Ringle et al., 2012; Wilson, 2010), PLS-SEM-specific data segmentation techniques (e.g., Rigdon et al., 2010; Sarstedt, 2008), analysis of interaction effects (e.g., Henseler and Chin, 2010; Henseler and Fassott, 2010) and other nonlinear effects (Dijkstra and Henseler, 2011; Henseler et al., 2012a; Rigdon et al., 2010) or multigroup analysis procedures (e.g., Chin and Dibbern, 2010; Sarstedt et al., 2011b).

A summary of the main findings of recent reviews of PLS-SEM applications across a variety of disciplines suggests that the following issues are prevalent and require the particular attention of authors, reviewers, and editors to improve the quality of PLS-SEM studies:

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- *Goal of the analysis*

Some researchers claim they use PLS-SEM for predictive purposes. However, this aspect often is not reflected in the analysis and interpretation of results. For example, researchers who claim to use PLS-SEM for their study's predictive purpose frequently accept relatively low levels of R^2 values and/or do not analyze criteria such as the predictive relevance Q^2 and related measures (Hair et al., 2013; Henseler et al., 2012b).

- *Small sample sizes*

PLS-SEM is advantageous when used with small sample sizes (e.g., in terms of the robustness of estimations and statistical power; Reinartz et al., 2009). However, some researchers abuse this advantage by relying on extremely small samples relative to the underlying population. All else being equal, the more heterogeneous the population in a structure is the more observations are needed to reach an acceptable sampling error level. Ignoring the fundamentals of sampling theory yields meaningless results no matter which method is applied. PLS-SEM has an erroneous reputation for offering special sampling capabilities that no other multivariate analysis tool has. Like any other statistical technique, inference statistics based on PLS-SEM require representative samples. Researchers are therefore well-advised to use sampling techniques carefully (which also applies to all other studies using different analysis techniques) and carefully consider the statistical power of their analyses (see Hair et al., 2013 for an overview).

- *Reporting of data characteristics and missing data*

Similarly, researcher frequently claim they use PLS-SEM because their data is non-normal. Additional descriptive statistics (e.g., skewness and kurtosis) to substantiate this argument and to inform the reader about the data are routinely omitted, even though they have important implications for the analyses (e.g., with regard to the use of bootstrapping procedures; Hair et al., 2013). Statements about missing values are also rare, as hardly any studies provide information on the magnitude of the missing data (e.g., percentages per variable and case) and the missing value treatment option used (e.g., mean value replacement or case-wise deletion). Providing information on missing values and potential non-response bias is crucial to maximize confidence in the results. Furthermore, treatment of missing values has vast implications for PLS-SEM use. For example, even if a relatively small number of missing values has been replaced by the mean value of the remaining (valid) observations, methods for analyzing unobserved heterogeneity (e.g., by using FIMIX-PLS; Hahn et al., 2002; Rigdon et al., 2011; Ringle et al., 2010a,b; Sarstedt et al., 2011a; Sarstedt and Ringle, 2010) will yield highly biased results. Researchers need to fully report data characteristics and treat missing values carefully in their analyses.

- *Using categorical data*

PLS-SEM software applications, such as SmartPLS (Ringle et al., 2005), provide results for all types of variables, regardless of whether they have metric, quasi-metric, ordinal, or categorical scales (e.g., binary coded). In its basic form, however, PLS-SEM requires metric or quasi-metric data. The use of other scale levels changes the interpretation of results or violates the method's fundamental requirements.¹ For example, one should not use dummy variables in reflective measurement models. Moreover, the use of dummy variables in formative measurement models requires an interpretation similar to that for regression analyses with dummy variables (Hair et al., 2010). Thus, researchers using PLS-SEM in such data constellations should be acquainted

¹ Basically, PLS-SEM builds on the ordinary least squares method, which requires the dependent variable to be continuous. If it is not, this requirement is violated when estimating a PLS path model, which has significant consequences for the analysis.

with the basic principles of the PLS-SEM algorithm to understand which model set-ups allow for the use of categorical variables.

- *Formative measurement models*

Researchers often use PLS-SEM because this technique facilitates the inclusion of formative measurement models, which have recently attracted considerable attention across various disciplines (e.g., Diamantopoulos and Riefler, 2011; Diamantopoulos et al., 2008; Diamantopoulos and Winklhofer, 2001; Gudergan et al., 2008; Jarvis et al., 2003; MacKenzie et al., 2005; MacKenzie et al., 2011). However, formative measurement models should be evaluated cautiously, for instance, by following the recommendations given by Hair et al. (2013) in the context of PLS-SEM. Like reflective measurement models, they require a thorough evaluation, for example, by assessing the indicator weights, the loadings, or by carrying out redundancy analyses.² If formative measurement models are not carefully evaluated the value of the entire analysis is in doubt. Finally, the CTA-PLS analysis (Gudergan et al., 2008) allows distinguishing empirically a formative measurement model specification from a reflective one. All these aspects are usually not taken into account when researchers include formative measurement models in their PLS-SEM study.

- *Control variables*

Control variables are often included in PLS path models, accounting for some of the target construct's variation. Regardless of whether these control variables are significant or not, the results for control variables are usually not further interpreted. When the effect of control variables are significant, the researcher should use this finding especially carefully when drawing conclusions or initiating additional analyses (e.g., PLS-SEM multigroup analyses; Sarstedt et al., 2011b), which occurs routinely with regression-based models (e.g., Raithel et al., 2012). Lastly, researchers usually neglect the fact that the inclusion of control variables increases model complexity and, thus, may also increase the required sample size for estimating the PLS path model.

- *Moderator analysis*

Two crucial issues often lead to problems in moderator analysis. First, the moderators are included with their interaction terms (i.e., the multiplication of indicators or constructs) in the PLS path model and their simple effects are mistakenly interpreted as main effects (Jaccard and Turrisi, 2003). The character of these results differs completely, however, and requires particular attention when analyzing PLS-SEM results (Hair et al., 2013; Henseler and Chin, 2010; Henseler and Fassott, 2010).³ One may first estimate and evaluate the main effects in the PLS path model and, in a subsequent moderator analysis, include the product term and its interaction effect to avoid the common mistake of confounding main and simple effects. As an additional remedy, researchers should consider using orthogonalization (Henseler and Chin, 2010). Second, when there is theoretical support for multiple moderators,

² Note that standard assessment procedures for formative measurement models disregard the loadings. Recent research however (Cenfetelli and Bassellier, 2009; Hair et al., 2013) suggests that in situations in which the weights are not significant, researchers should also consider the indicators' absolute contribution to the construct (i.e., the loadings).

³ A PLS path model without moderation includes only main effects between latent variables in the structural model. The main effects model becomes a moderator model after including a product term and its interaction (or moderating) effect. In a moderator model, main effects change into simple (or single) effects (Henseler and Fassott, 2010). This distinction is important and must be properly accounted for when interpreting PLS-SEM results. Whereas a main effect quantifies the change in the level of the dependent variable when the considered independent variable is increased by one unit and all other independent variables remain constant (*ceteris paribus*), a simple effect quantifies the change in the level of the dependent variable when the independent variable is increased by one unit, the interacting variable has a value of zero, and all other independent variables remain constant. Henseler and Fassott (2010) and Hair et al. (2013) provide detailed explanations and examples on testing moderating effects in PLS-SEM.

one may consider analyzing one moderator a time to maintain interpretability of all results. Researchers often focus simultaneously on multiple moderators. However, the question remains whether the interactions between multiple moderators (e.g., age and income) should be included. Even if such interactions are accounted for, the results are usually not properly interpreted (e.g., does the moderator age affect the interaction effect of the moderator income and how does this three-way interaction change the interpretation of the moderator model estimates?).

- *Non-linear models*

The ever-increasing complexity of theories and model constellations has increased the prominence of the modeling of non-linear effects (Cortina, 1993). While nonlinear effects can be easily included in a PLS path model (Dijkstra and Henseler, 2011; Henseler et al., 2012a; Rigdon et al., 2010), their actual use is a matter of concern. First, researchers mostly do not provide a precise a priori reasoning for the shape of the non-linear effect. Thereby, they do not substantiate their hypotheses. Merely searching for a nonlinear effect entails capitalizing on chance results. The more you search in many different directions, the higher the probability you will find something. However, it is very unlikely that any replication of such an ex post facto analysis would yield exactly the same effect (Cliff, 1983). Consequently, the corresponding outcomes are not generalizable. Second, even though nonlinear effects are frequently significant, the question remains regarding their added value. A critical look at the outcomes often indicates significant but relatively small sizes of the quadratic effect, entailing almost no changes in the R^2 values. In such cases, a linear and, thus simpler, model (i.e., more parsimonious) provides almost the same results. In the tradition of Ockham's razor, researchers usually opt for the more parsimonious approach if admissible. Third, the interpretation of nonlinear effects requires the researcher to characterize the different fractions of a function. Consider, for example, a u-shaped relationship depending on the size of firms: the fraction with a strong negative slope for smaller firms, the fraction with a slope of approximately zero for mid-sized firms, and the fraction with a strong positive slope for large firms. The different fractions may be explained by the source of the nonlinear effect itself. For this purpose, a PLS-SEM multigroup analysis (Chin and Dibbern, 2010; Sarstedt et al., 2011b) can be used to approximate the different fractions of a non-linear relationship through group-specific linear regressions with different slope sizes and/or signs. Again, this approach allows the researcher to switch from the more complex non-linear design to the more parsimonious linear model while including — as an additional explanation — the grouping of the data.

- *Mediator analysis*

Most structural models are subject to mediation effects (e.g., Hair et al., 2013; Helm et al., 2010), which many researchers overlook in their PLS-SEM analyses. In the extreme, they do not examine and interpret the result of a full mediation but simply state that a relationship between two latent variables is not significant. Hence, they erroneously conclude that, in the structural model, the relationship between the two latent variables is nil. Researchers should routinely report the total effects (i.e., the sum of direct and indirect effects between two constructs; Albers, 2010). This not only allows a more complete picture of the mediating constructs' role, but also provides practitioners with actionable results regarding cause-effect relationships. Moreover, formalized mediation analysis by means of bootstrapping analysis (Hair et al., 2013) is especially valuable when corresponding hypotheses have been formulated (e.g., Sattler et al., 2010). A final note of caution refers to the quality of measurement models in the context of mediation. When using variance-based structural equation modeling techniques such as PLS-

SEM, researchers must strive for a very reliable measurement of the mediator variable (otherwise, wrong conclusions might be drawn; [Henseler, 2012](#)).

- *Hierarchical component models or higher (e.g., second) order models*

Hierarchical component models are relatively easy to conduct in PLS-SEM by using [Wold's \(1982\)](#) repeated indicator approach. Recent explications provide additional insights into how hierarchical component models should be incorporated into PLS path models ([Becker et al., 2012](#); [Hair et al., 2013](#); [Ringle et al., 2012](#); [Wetzels et al., 2009](#)). Unfortunately, users often misapply higher order models both conceptually and technically. Instead of presenting proper explications that justify the type of higher order model used (e.g., the reflective-reflective type or the reflective-formative type), researchers routinely use hierarchical component models to summarize information in a second, third or even higher dimension of abstraction with the plausible objective of reducing the number of relationships in the PLS path model. The resulting higher order models however are often difficult to defend from a theoretical point of view.

Despite its ease of application, the repeated indicator approach may entail several pitfalls. First, the number of indicators per lower order component should be balanced ([Becker et al., 2012](#)). Otherwise the estimated relationships between the higher and the lower order components may be biased. Second, researchers usually do not evaluate the higher order constructs, although the same evaluation criteria (and their critical values) used for the lower order components apply. Hence, information about relevant evaluation criteria outcomes is important and should be provided. Third, some researchers include relationships from other latent variables in the structural model, which are not part of the hierarchical component model, to formative higher order constructs. These relationships always have a value of approximately zero when the indicator reuse technique is applied to determine the higher order construct in PLS-SEM, because the formative lower order components already explain all of the former's variance. Hence, the conclusion that other constructs in the structural model do not explain any variation of the reflective-formative type or the formative-formative type higher order construct would be false and misleading, as this is a technical outcome of the repeated indicator approach. In such situations, a two-stage approach ([Ringle et al., 2012](#)) should be used, which allows for handling this technical limitation of the repeated indicator approach.

- *Results evaluation and reporting*

The results evaluation and reporting are often incomplete. In some instances, the path coefficients and their significance are not reported, or the R^2 values of the endogenous latent variables are missing. Even if a study shows all the relevant criteria as suggested, for example, by [Hair et al. \(2013\)](#), the researchers often report the results without critical reflection or further interpretation. A fundamental issue relates to the use of the Goodness-of-Fit (GoF) index proposed by [Tenenhaus et al. \(2004, 2005\)](#) as a means to validate a PLS path model globally. [Henseler and Sarstedt \(2013\)](#) challenged the usefulness of the GoF index conceptually and empirically. For instance, in a simulation study, the authors show that the GoF index cannot separate valid models from invalid ones. Moreover, the GoF index is not applicable to formative measurement models and does not penalize overparameterization efforts. Furthermore, the software used (for a PLS software comparison see [Temme et al., 2010](#)) is usually not reported, although this information provides important details regarding the default values used in the running of the PLS-SEM algorithm and supplementary analyses. When authors do provide this important information, they usually fail to correctly cite the two major software applications SmartPLS ([Ringle et al., 2005](#)) and PLS-Graph ([Chin, 2003](#)) as required in these software's license agreements (i.e., more than 50 percent of PLS-SEM applications are subject to this flaw).

Like all statistical methods, “PLS-SEM requires several choices that, if not made correctly, can lead to improper findings, interpretations, and conclusions” (Hair et al., 2012c, p. 415). PLS-SEM analyses should account for the issues raised by Gefen et al. (2011) regarding structural equation modeling in general, as well as those reported by Hair et al. (2012a,b,c) regarding the PLS-SEM method in particular.⁴ These requirements can be grouped into the following key categories (Hair et al., 2013; Hair et al., 2011; Hair et al., 2012b,c):

1. Data characteristics
 - Assess the required sample size for estimating the established PLS path model to ensure a sufficient level of statistical power,
 - Provide a comprehensive description of the sample (e.g., report the number of observations and amount of missing values but also mean values, variances and other descriptive statistics),
 - Characterize the distribution of the variables (e.g., report skewness and kurtosis of data),
 - Use a holdout sample (e.g., conduct the analysis with 70% of the original sample),
 - Offer all relevant information to facilitate replication of your analysis (e.g., share the correlation/covariance matrix or raw data in an online appendix),
 - Explain in detail the scales of variables; treat variables with scales other than metric or quasi-metric with particular care in PLS-SEM (e.g., do not use categorical variables in endogenous constructs; interpret categorical variables in exogenous constructs carefully).
2. Model characteristics
 - Fully describe the structural model (i.e., the latent variables and their relationships) using a graphical (instead of a formal/mathematical) representation of the PLS path model, which makes it much easier for the reader to quickly grasp some of the key research contents.
 - Characterize the measurement models (formative vs. reflective) of the latent variables (e.g., empirically substantiate the selected measurement model by using CTA-PLS) and include a complete list of the indicators employed in the measurement models (e.g., in the appendix).
3. PLS algorithm settings and software used by reporting...
 - ...the starting values of the weights for the initial approximation of the latent variable scores (e.g., use a uniform value of 1 as an initial value for each of the outer weights),
 - ...the inner weighting scheme to determine the latent variable scores,
 - ...the stop criterion (e.g., the sum of the measurement model weights' absolute changes between two iterations < 0.00001),
 - ...the actual number of iterations (e.g., 300),
 - ...the software used (e.g., report the software, including the version to indicate the default settings).
4. Report the exact algorithm settings used when running supplementary procedures (e.g., bootstrapping, blindfolding, CTA-PLS, analysis of moderating effects, multi-group analysis, FIMIX-PLS) in detail.
5. Conduct a comprehensive evaluation of the results...
 - ...of reflective measurement models by including the following
 - Indicator reliability (e.g., standardized indicator loadings ≥ 0.70 ; in exploratory studies, loadings of 0.40 are acceptable),

⁴ Besides the guidelines given in this editorial, one finds various recommendations on the use of the PLS-SEM method and the evaluation of results in prior literature; examples include: Chin (1998, 2010); Falk and Miller (1992); Götz et al. (2010); Haenlein and Kaplan (2004); Hair et al. (2011); Hair et al. (2012); Henseler et al. (2009); Lohmöller (1989); Roldán and Sánchez-Franco (2012); Sosik et al. (2009); Tenenhaus et al. (2005).

- Internal consistency reliability: Use Cronbach's alpha as the lower bound of the internal consistency reliability and composite reliability as the upper bound for the (unknown) true reliability. Both measures should exceed 0.70,
 - Convergent validity (e.g., $AVE \geq 0.50$),
 - Discriminant validity (e.g., report and analyze Fornell-Larcker criterion results and/or cross loadings),
 - ...of formative measurement models by including
 - Indicators' relative contribution to the construct (i.e., the indicator weights),
 - Significance of weights (e.g., report t-values, p-values or standard errors),
 - Indicators' absolute contribution to the construct (i.e., the indicator loadings), depending on the results of the prior analyses,
 - Multicollinearity assessment (e.g., $VIF < 5$; tolerance > 0.20),
 - Redundancy analyses (Hair et al., 2013).
 - ...of the structural model
 - R^2 (e.g., an acceptable level depends on the research context),
 - Effect size f^2 (e.g., 0.02, 0.15, 0.35 for weak, moderate, strong effects),
 - Path coefficient estimates (e.g., use bootstrapping to assess significance; provide confidence intervals),
 - Predictive relevance Q^2 and q^2 (e.g., use blindfolding; $Q^2 > 0$ is indicative of predictive relevance; q^2 : 0.02, 0.15, 0.35 for weak, moderate, strong degree of predictive relevance of each effect),
 - Analyze observed and unobserved heterogeneity (e.g., consider categorical or continuous moderating variables using a priori information or FIMIX-PLS).
6. Conduct complementary PLS-SEM analyses (e.g., mediator, moderator, multi-group or importance-performance analyses) in subsequent steps after the analyses of the original PLS path model as described in the previous steps.

These guidelines, as well as new textbooks on the PLS-SEM method (e.g., Hair et al., 2013) provide researchers, editors, and reviewers with the knowledge they need to ensure the rigor of published research in academic journals. By following these recommendations, the quality of studies employing PLS-SEM should be enhanced and crucial mistakes avoided. As in any empirical research, the goal in PLS-SEM is to progress towards the highest possible level of transparency that allows the replication of published studies. Future developments towards this goal will substantially improve the way in which research is conducted, as well as the quality of published articles.

This second *Long Range Planning* (<http://www.journals.elsevier.com/long-range-planning/>) special issue on PLS-SEM in strategic management research and practice seeks to further progress towards this goal. The journal received 41 articles for its special issue on PLS-SEM, twelve of which completed a thorough review process successfully. Based on the number of high quality manuscripts, a decision was made to split the special issue. In the first *Long Range Planning* special issue on PLS-SEM in strategic management (Hair et al., 2012a; Robins, 2012), the focus was on methodological developments and their application (Becker et al., 2012; Furrer et al., 2012; Gudergan et al., 2012; Hair et al., 2012a,b,c; Money et al., 2012; Rigdon, 2012). This second special issue provides a forum for topical issues that demonstrate the usefulness of PLS-SEM by piloting applications of this method in the field of strategic management with strong implications for strategic research and practice. As such, the special issue targets two audiences: academics involved in the fields of strategy and management, and practitioners such as consultants. The six articles in this issue are summarized in the following paragraphs.

In their paper "Crossing Borders and Industry Sectors: Behavioral Governance in Strategic Alliances and Product Innovation for Competitive Advantage," Yong Kyu Lew and Rudolf R. Sinkovics investigate how international technology alliances (ITAs) between software and hardware firms in

the mobile computing market, involving major players such as Motorola, HTC, Samsung, Apple, etc., benefit from behavioral governance mechanisms. The empirical analysis is based on a cross-industry and cross-border PLS path model which identifies technological commitment as a key factor in expediting technology resource exchange in ITAs. Technological commitment — that is, honest efforts to deliver on promises to the partner and willingness to make investments in support of their partners — contributes to firm-level innovation capabilities, and increased performance outcomes. The results also show that firm-level performance is only influenced by market development capability, and not new product development capability. This comes as a surprise to innovation researchers, but highlights the overarching importance of functional marketing. This paper is an important contribution to a better understanding of how high-tech firms benefit from relational inter-firm governance in international technology resource exchange arrangements. Furthermore, it provides evidence of the methodological usefulness of PLS-SEM in strategic alliance, capability, and performance research.

The next paper is by Liselore Berghman, Paul Matthyssens, Sandra Streukens, and Koen Vandenbempt and titled “Deliberate learning mechanisms for stimulating strategic innovation capacity.” Their notable and piloting PLS-SEM study reveals that strategic innovation capacity is directly or indirectly strengthened when managers deliberately install specific learning mechanisms for the three dimensions of absorptive capacity: knowledge recognition, assimilation and exploitation. Their analysis of the antecedents and outcomes complements and extends existing research on absorptive capacity by indicating the importance of deliberate action when trying to break through existing industry practices.

In the third paper, “Dynamic Capabilities and Performance: Strategy, Structure and Environment,” Ralf Wilden, Siegfried Gudergan, Bo Nielsen, and Ian Lings argue theoretically and demonstrate empirically that the performance effects of dynamic capabilities are contingent on organizational structure and the competitive intensity in the market. Their PLS-SEM analysis shows that organic organizational structures facilitate the impact of dynamic capabilities on organizational performance. Furthermore, their research identifies that the performance outcomes of dynamic capabilities are contingent upon the competitive intensity that firms face. Their study provides evidence of the performance effects of internal alignment between organizational structure and dynamic capabilities, as well as the external fit of dynamic capabilities with competitive intensity. Finally, the advantages of PLS-SEM for modeling latent constructs, such as dynamic capabilities, are summarized and relevant managerial implications provided. In doing so, the authors provide a sophisticated approach to measuring dynamic capabilities.

The next paper in this special issue by Christian Landau and Carolin Bock — “Value Creation through Vertical Intervention of Corporate Centers in Single Business Units of Unrelated Diversified Portfolios — The Case of Private Equity Firms” — addresses the value creation by multi-business firms’ corporate centers through vertical intervention in single business units. Drawing on parenting literature, agency theory, and the resource-based view, the authors develop a model comprised of an interlinked set of hypotheses on value creation. The results of their PLS path model estimation show that corporate centers can create value through vertical intervention in single business units by, as part of their administrative function, reducing agency costs through governance mechanisms and, interestingly, also providing businesses access to strategic resources during the course of their entrepreneurial function. In particular, the provision of intangible resources — that is, the knowledge and networks of the corporate center — has a value creating effect on business units, and suggests a new direction for future research in this field.

The fifth paper is “Why Social Currency becomes a Key Driver of a Firm’s Brand Equity — Insights from the Automotive Industry” by Lara Lobschat, Markus A. Zinnbauer, Florian Pallas, and Erich Joachimsthaler. This article conceptualizes a new multifaceted formative construct, social currency, and its dimensions based on social capital theory to capture the complex social nature of brands. Social currency is defined as the value that is accumulated by customers communicating, interacting and, thereby, spreading brand-related information to other customers. The PLS-SEM analysis with data from a representative U.S. consumer survey in the automotive context then

empirically validates their construct and its multiple dimensions, and substantiates its nomological validity by exploring its influence on well-established brand equity measures. Their results confirm a valid description of social currency and demonstrate a positive effect of social currency on the brand equity measures of perceived quality, brand loyalty, and brand trust. Moreover, their findings provide researchers and managers with crucial insights on how to assess social currency.

The final paper of this special issue, “A Model of Antecedents and Consequences of Intuition in Strategic Decision Making: Evidence from Egypt,” is by Said Elbanna, John Child, and Mumin Dayan. It advances our understanding of decision making by proposing and examining a model of the antecedents and consequences of intuition in strategic decision-making. The conceptualization of intuition in managerial decision-making is a ground-breaking effort for the future study of this emerging area. Moreover, the authors’ study of Egyptian manufacturing firms indicates that decision uncertainty and company size are related to the use of intuition; that intuition significantly influences decision disturbance; and that environmental hostility moderates the relationship between decision intuition and disturbance. Finally, the paper summarizes the implications of these findings for strategic decision-making theory and practice, as well as for further research.

Finally, we would like to use this opportunity to draw your attention to our acknowledgement in the editorial of the first special issue on PLS-SEM in strategic management (Hair et al., 2012a). Without the extraordinary support of the reviewers, who contributed their valuable time and talent to develop this special issue, and ensured the articles’ quality with their constructive comments and suggestions to the authors, this special issue would not have been possible. Finally, we would like to take this opportunity to thank the *Long Range Planning* Editor-in-Chief, James A. Robins (Vienna University of Economics and Business), for initiating this project and his thorough assistance — with the superb support of the journal’s editorial office — in developing this special issue. We really enjoyed working with highly committed authors and reviewers for a well-organized top-tier journal. Thanks again, and continue your excellent work!

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